DAA LAB 6

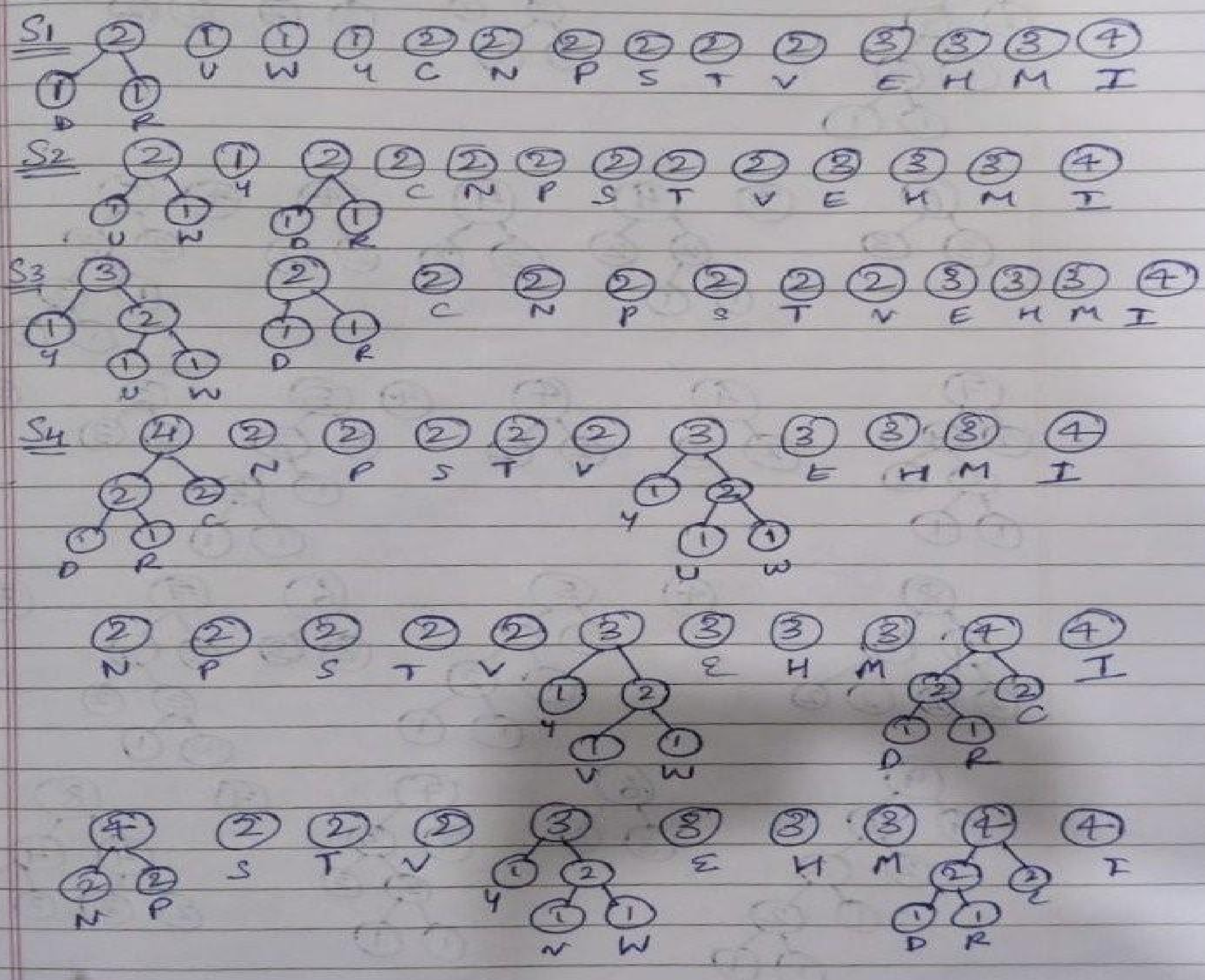
Using Huffmann coding, arrive max encoding bits for phrase

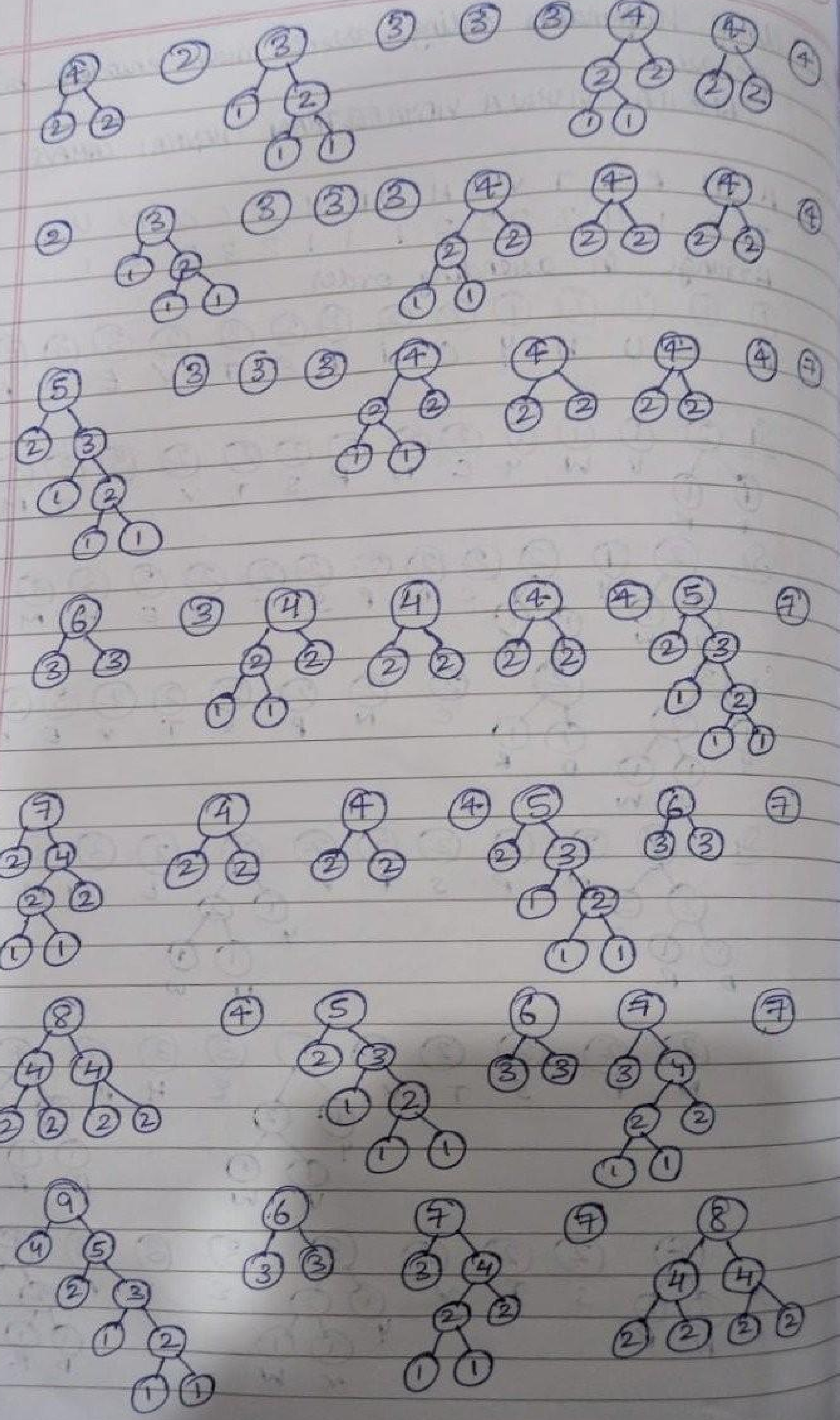
AMRITA VISHWA VIDYAPEETHAM CHENNAI CAMPUS

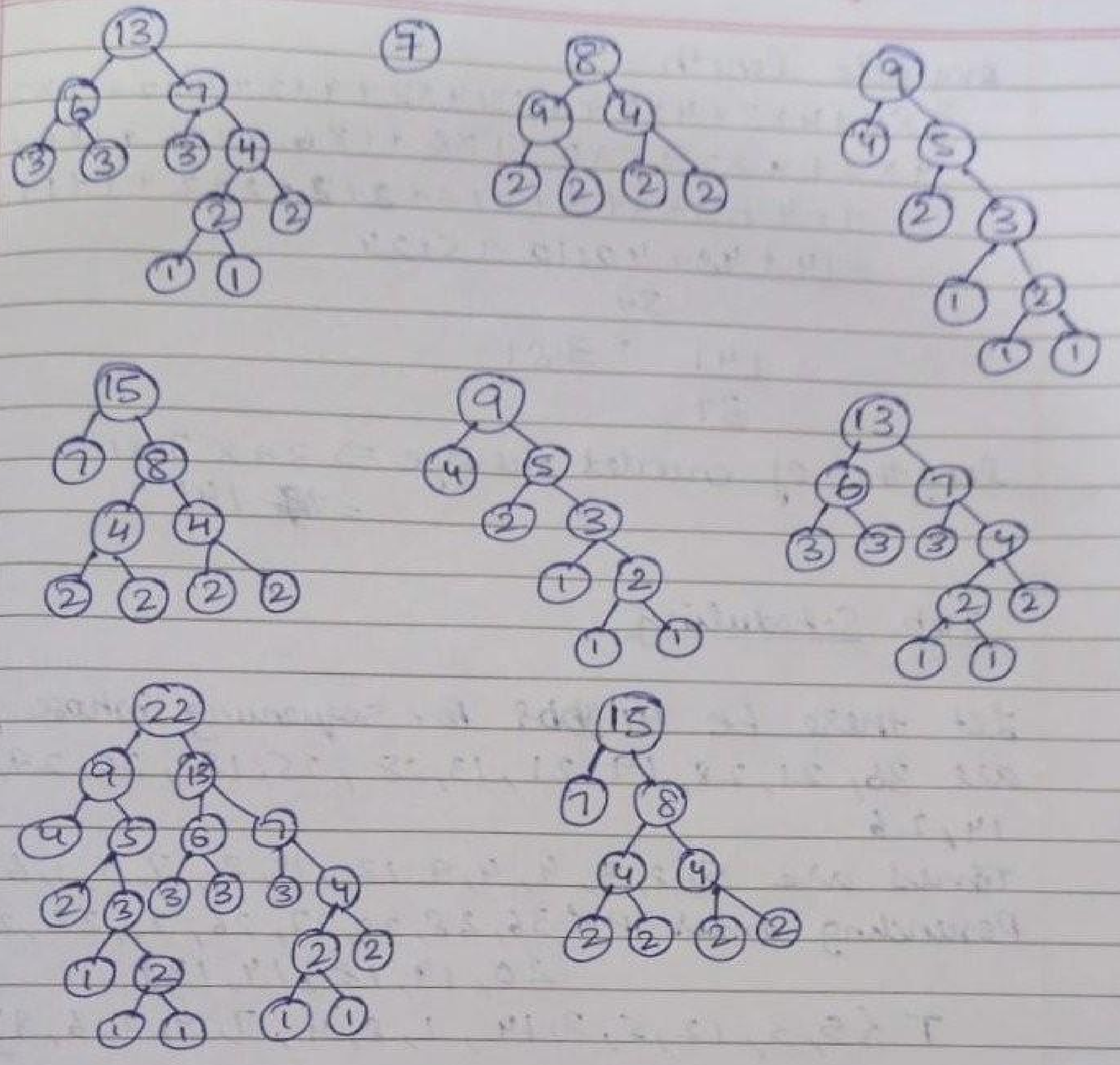
A	M	R	I	T	V	S	H	W	D	Y	P	E	C	N	U
7	3	1	4	2	2	2	3	1	1	1	2	3	2	2	1

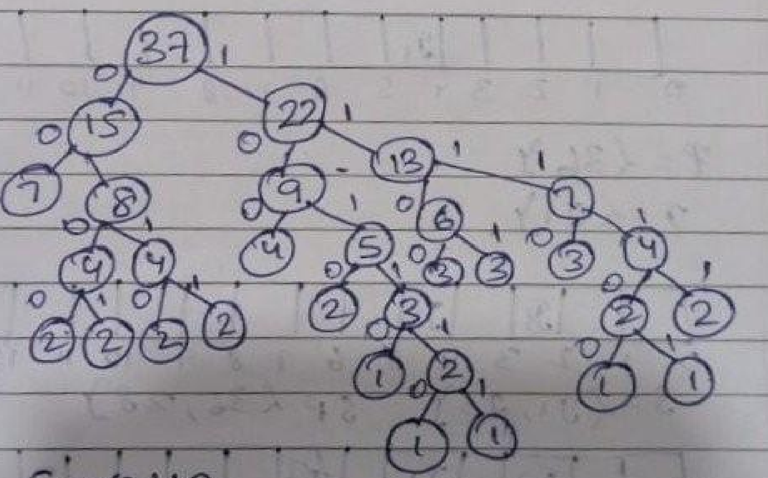
Arrange in ascending order

1	1	1	1	1	2	2	2	2	2	2	3	3	3	4
D	R	U	W	Y	C	N	P	S	T	V	E	H	M	I









A - 00
I - 100
V - 1010
E - 1100
H - 1101
M - 1110
N - 0100
P - 0101

S - 0110
T - 0111
Y - 10110
U - 101110
W - 101111

D - 111100
R - 111101
C - 11111

Avg code length

$= \frac{7\times2+4\times3+4\times3+4\times3+4\times3+4\times2+4\times2+4\times2+4\times2+4\times2+5\times2+1\times5+1\times6+1\times6+1\times6+1\times6}{7+4+3+3+3+2+2+2+2+2+2+1+1+1+1+1}$

$= \frac{14+48+40+10+5+24}{37}$

$= \frac{141}{37} = 3.81$

length of encoded message $\Rightarrow 37\times3.81$

$= 141$

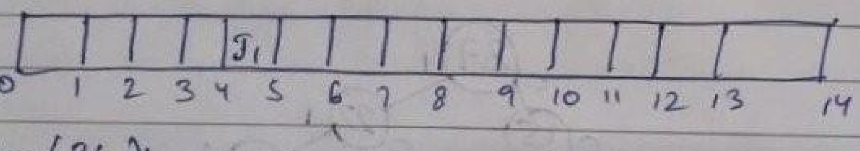
Job Scheduling

Let there be 14 jobs for sequencing whose profits are 36, 21, 28, 19, 21, 13, 28, 25, 18, 20, 27, 22, 14, 26

Times are 5, 8, 3, 4, 4, 9, 12, 14, 2, 7, 5, 1, 6, 3

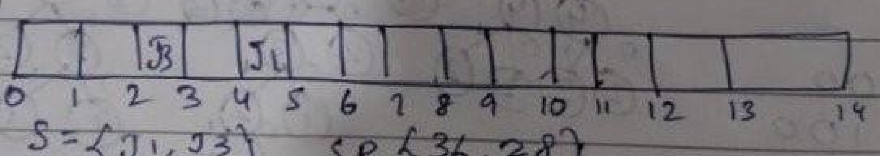
Descending order P <36, 28, 28, 27, 26, 25, 22, 21, 21, 20, 19, 18, 14, 13}

T {5, 3, 12, 5, 3, 14, 1, 8, 4, 7, 9, 2, 6, 9}

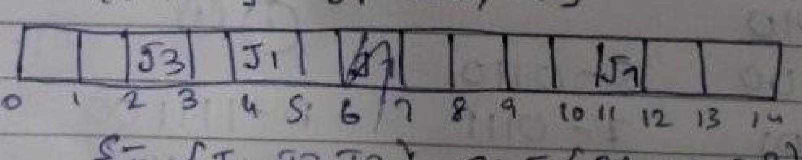


P = {36}

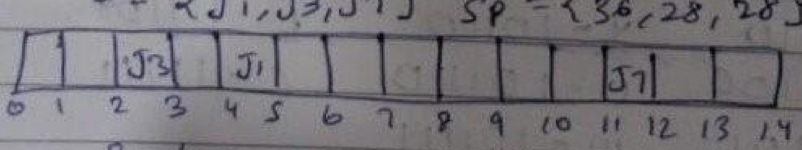
J = {J1}



S = {J1, J3} SP {36, 28}



S = {J1, J3, J7} SP = {36, 28, 28}



S = {J1, J3, J7, J11} Sp = {36, 28, 28, ~~27~~}

		J3	J1							J7		J8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

S = {J1, J3, J7, J8} SP = {36, 28, 28, 25}

J12		J3	J1		J2				J7		J8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

S = {J1, J3, J7, J8, J12, J2} SP = {36, 28, 28, 25, 22, 21}

J12		J3	J5	J1		J2			J7		J8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

S = {J1, J3, J7, J8, J8, J12, J2, J5} SP = {36, 28, 28, 25, 22, 21, 21}

J12		J3	J5	J1	J10	J2			J7		J8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

S = {J1, J3, J7, J8, J12, J2, J5, J10} SP = {36, 28, 28, 25, 22, 21, 21, 20}

J12	J9	J3	J5	J1	J10	J2			J7		J8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

S = {J1, J3, J7, J8, J12, J2, J5, J10, J9}

J12	J9	J3	J5	J1	J13	J10	J2			J7	J8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

S = {J1, J3, J7, J8, J12, J2, J5, J10, J9, J13}

J12	J9	J3	J5	J1	J13	J10	J2	J6			J7	J8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

S = {J1, J3, J7, J8, J12, J2, J5, J10, J9, J13, J6}

0 1 2 3 4 5 6 7 8 9 10 11 ~~12 13 14~~

S = {J1, J3, J7, J8, J12, J2, J5, J10, J9, J13, J6}

SP = {36, 28, 28, 25, 22, 21, 21, 20, 18, 14, 13}

DAA LAB 6

```c
#include <stdio.h>
#include <stdlib.h>
#define MAX 100
struct Node {
    char data;
    int freq;
    struct Node *left, *right;
};
struct Node* createNode(char data, int freq) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->freq = freq;
    newNode->left = newNode->right = NULL;
    return newNode;
}
void printCodes(struct Node* root, int arr[], int top) {
    if (root->left) {
        arr[top] = 0;
        printCodes(root->left, arr, top + 1);
    }

    if (root->right) {
        arr[top] = 1;
        printCodes(root->right, arr, top + 1);
    }

    if (!root->left && !root->right) {
        printf("%c: ", root->data);
        for (int i = 0; i < top; i++)
            printf("%d", arr[i]);
        printf("\n");
    }
}
int main() {
    char data[] = {'A', 'B', 'C', 'D'};
    int freq[] = {5, 9, 12, 13};
    int size = 4;
    struct Node *n1 = createNode(data[0], freq[0]);
    struct Node *n2 = createNode(data[1], freq[1]);
    struct Node *n3 = createNode(data[2], freq[2]);
    struct Node *n4 = createNode(data[3], freq[3]);
    struct Node *combine1 = createNode('$', n1->freq + n2->freq);
    combine1->left = n1;
    combine1->right = n2;
    struct Node *combine2 = createNode('$', n3->freq + n4->freq);
    combine2->left = n3;
    combine2->right = n4;
    struct Node *root = createNode('$', combine1->freq + combine2->freq);
    root->left = combine1;
    root->right = combine2;
    int arr[MAX], top = 0;
    printf("Huffman Codes:\n");
    printCodes(root, arr, top);
    return 0;
}
```

```
sailakshmi@LAPTOP-GUU0DK01:~$ cd "/mnt/c/Users/Sai Lakshmi Tejasri/OneDrive/Desktop/c programs daa"
sailakshmi@LAPTOP-GUU0DK01:/mnt/c/Users/Sai Lakshmi Tejasri/OneDrive/Desktop/c programs daa$ gcc huffman.c -o huffman
sailakshmi@LAPTOP-GUU0DK01:/mnt/c/Users/Sai Lakshmi Tejasri/OneDrive/Desktop/c programs daa$ ./huffman
Huffman Codes:
A: 00
B: 01
C: 10
D: 11
```

DAA LAB 6

```
#include <stdio.h>
struct Job {
    int id;
    int deadline;
    int profit;
};
void jobScheduling(struct Job jobs[], int n) {
    int i, j;

    // Sort jobs by profit (descending)
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (jobs[i].profit < jobs[j].profit) {
                struct Job temp = jobs[i];
                jobs[i] = jobs[j];
                jobs[j] = temp;
            }
        }
    }
    int slot[10] = {0};
    int totalProfit = 0;
    printf("Selected Jobs: ");
    for (i = 0; i < n; i++) {
        for (j = jobs[i].deadline - 1; j >= 0; j--) {
            if (slot[j] == 0) {
                slot[j] = jobs[i].id;
                totalProfit += jobs[i].profit;
                printf("J%d ", jobs[i].id);
                break;
            }
        }
    }
    printf("\nTotal Profit: %d\n", totalProfit);
}
int main() {
    struct Job jobs[] = {
        {1, 2, 100},
        {2, 1, 19},
        {3, 2, 27},
        {4, 1, 25},
        {5, 3, 15}
    };
    int n = 5;
    jobScheduling(jobs, n);
    return 0;
}
```

```
sailakshmi@LAPTOP-GUU0DK01:/mnt/c/Users/Sai Lakshmi Tejasri/OneDrive/Desktop/c programs daa$ gcc jobscheduling.c -o jobscheduling
sailakshmi@LAPTOP-GUU0DK01:/mnt/c/Users/Sai Lakshmi Tejasri/OneDrive/Desktop/c programs daa$ ./jobscheduling
Selected Jobs: J1 J3 J5
Total Profit: 142
```